



INDUSTRIAL MANAGEMENT AND SAFETY

MODULE 1



❖ LESSON1-PRINCIPLES OF MANAGEMENT

Any organization where a group of persons joins together for the attainment of some objective or objectives whatever may be the nature and kind of the organization, it cannot run successfully unless there is someone to manage its affairs. Management is an essential part of any group activity.

Management

- Management is a system to direct the activities of the organization so that human and material resources may be effectively used to achieve desired objectives. Management also defined as the art of getting work done efficiently through and by the people, within satisfaction for employer, employees and the public.
- Actually Management is the process of Managing *Men, Materials, Money, Machinery, Market, and Methods*. i.e., 6M's.
- If the workers are not willing to work hard, our whole projects will be failure. So the managing men is the most important factor to be considered.
- In this context the management can be defined as creating the environment of an enterprise where individuals working together in groups can perform efficiently and effectively towards the attainment of group goals.
- Management should act as a creative as well as a motivating force in the organization. It should ensure a smooth flow of work in the organization by focusing on strong points, neutralizing weak links, overcoming difficulties and establishing a team spirit.

Characteristics of management

1. It should coordinate all the people in an organization for achieving their goals.
2. It should utilize all the available resources and secure the maximum result.
3. It should distribute the work among the workers based on their skill and effort
4. It should accept and encourage any new ideas from the personnel of the industry that may result in better production
5. It should motivate people and provide training for the workers to improve their skills.
6. It should have better inspection over the workers for their output
7. It has to maintain discipline and to keep control over the employees of various sections
8. It has to make the arrangement of payments and their records etc.
9. By adhering to high standard of performance it must be able to provide facilities and wages to the workers and reduce the prices.

10. It should look after whether the target production has been achieved as per market demand

Development of management theory

- The story of management is essentially the story of mankind.
- The Egyptian pyramids and the Great Wall of China are current evidence that projects of tremendous scope, employing tens of thousands of people, were undertaken well before modern times.
- Without proper planning, organizing and controlling such monumental projects would not have been completed satisfactorily.
- The early management pioneers such as Robert Owen, Charles Babbage, Meta Calfe , Watt, Boulton and Henry Towne were provided solid mettle for constructing the sophisticated and elegant building of Management.
- A group of persons such as F.W. Taylor, Henry Fayol, H.C. Gantt, Emerson, Gilbert, Carl Barth and others contributed towards the process of the development of the science of Management.

There are nine concepts highlighting this period of management:-

1. Scientific management a key to productivity.
2. Decentralization for quicker decisions and better control.
3. Personnel Management as a means of selecting people into organization
4. Manager's development program to train managers how to meet the needs of tomorrow
5. Management accounting a foundation for managerial decision
6. Marketing
7. Long range planning
8. Role of operational research in management decision making
9. Financial Management for effective utilization of money resources in the organization.

Taylor's Scientific Management

Scientific management consists of the following:

- a) Observing and analyzing each task.
- b) Determination of a standard of work.
- c) Selecting and training men to perform their jobs.
- d) Ensuring that work is done in most efficient manner.

Frederick Winslow Taylor (F.W. Taylor) regarded as the father of scientific management emphasized the need for a systematic, rational, deliberate, scientific approach to all jobs and problems encountered in organized effort.

Instead of having each worker to do the job in his or her own way, Taylor sought to find the best way of doing the work.

Contributions of F.W. Taylor

1. He developed the principle of breaking a task (job) into elements for timing the same.
2. He kept himself involved in exploring the causes of inefficiency and labor difficulties in the industry. Through time studies he experimented to recognize losses of efficiency in Industrial operations.
3. He evolved certain principles of - Investigating work on scientific basis, selecting the best worker for a task and training him further to acquire desired skill, developing cooperative spirit between management and workers, almost equal division of work between workers and management, etc. which led to the concept of Scientific Management.
4. Another concept connected with the name of Taylor is A Fair Day's Task. While working on it, Taylor undertook studies on fatigue incurred by the workers and the time necessary to complete a task. Taylor suggested that for increasing production rate, the rate of each person should be planned at least one day in advance and he shall be allotted a definite work to complete by a given time using a pre-explained method.
5. Taylor developed functional organization in which one Foreman was made in charge for each function.
6. Taylor devoted his maximum attention towards time studies and he established work standards.
7. Taylor introduced and operated various costing systems.
8. Taylor suggested a wage incentive scheme known as Taylor's Differential piece rate plan.

Henry Fayol's Principles of management

Fayol known as the father of principles of management because he was first who suggested the functions of management. According to Fayol the activities of industrial organization could be divided into six categories:

1. Technical (Production and Manufacturing)
2. Commercial (Buying, Selling and Exchange)
3. Financial (Acquiring and using capital)

4. Security (Protection of Property and persons)
5. Accounting (Stock taking, Balance Sheet, Cost and Statistics)
6. Managerial (Planning, Organizing, Command, Coordination and control)

Henry Fayol's 14 Principles of management

1. Division of Work.
2. Authority and Responsibility.
3. Discipline.
4. Unity of Command.
5. Unity of Direction.
6. Subordination of individual's interest to general interest.
7. Remuneration of Personnel.
8. Centralization.
9. Scalar chain.
10. Order.
11. Equity.
12. Stability of Workers.
13. Initiative.
14. Espirite de corps

Comparison between the contributions of F.W.Taylor & Fayol

Similarities:-

1. Universality of management
2. Scientific methods
3. Importance of personnel
4. Improvement of practice
5. Idea through experience

Dissimilarities:-

1. Taylor for shop floor management, Fayol for top level management.
2. Taylor from bottom to upwards, Fayol from top to bottom.
3. Tylor for productivity, Fayol for theoretical approach.
4. Taylor for management, Fayol for administration.

FUNCTIONS OF MANAGEMENT

In every organisation managers required to perform certain Functions in order to achieve objectives/aims/goals. These functions are:

1. Planning
2. Organizing
3. Staffing
4. Directing
 - Leadership
 - Communication
 - Motivation
 - Supervision
5. Controlling
6. Decision making

1. Planning

Planning helps in determining the course of action to be adopted or achieving various organizational objectives. “It is a decision in advance i.e., what to do, when to do, how to do and who will do a particular job. Planning is a process which involves thinking before doing”. It involves the decision regarding organizational objective and developing policies, procedures. Programmes, budgets and strategies. Planning is a continuous process that works at all levels of management. A detailed planning is done in the beginning but the actual performance is reviewed and suitable changes are incorporated in plans when actual execution is done.

Plans may be of many types, such as short range plans, medium range plans. Standing plans, single use plans, strategic plans, administrative and operational plans.

Steps in Planning:

1. Collecting information
2. Laying down objectives
3. Developing premises
4. Examining alternative courses of action
5. Evaluation of working
6. Reviewing imitations of the system
7. Implementation of plans for achieving the targets.

2. Organizing

Organizing is defined as “the process of identifying and grouping work to be done, defining and delegating authority and responsibility, and establishing relationships for the purpose of enabling people to work most effectively together in accomplishing objectives”.

Steps in Organizing:

1. Identification of the work to be performed
2. To group the work of similar nature
3. To assign these groups of activities or work to individuals
4. To delegate authority and fix responsibility at various levels
5. To coordinate these authority responsibility relationship of various activities

3. Staffing

It is a process of recruiting, selecting and training the right people to fulfil the requirements of an enterprise. It is a continuous process, because new jobs may be created in the enterprise and existing employees may leave the concern.

Functions of staffing:

1. Manpower planning.
2. Recruitment, selection and training of manpower.
3. Placement of man power in requisite positions
4. Development, promotion, transfer and appraisal of manpower
5. Determination of employee remuneration and incentives

4. Directing

It is a process by which actual performance of subordinates is guided towards common goals of the enterprise. Direction is called management in action.

Functions of directing:

1. Issuing of orders and instructions to the subordinates
2. Guiding and teaching the subordinates the proper method of doing work, and
3. Supervising the subordinates to ensure that these work conforms to the plans

Directing involves the following:

- a) Leadership
- b) Communication
- c) Motivation
- d) Supervision

5. Controlling

It is the process that measures the current performance and guides it towards some predetermined goals. It is necessary in order to ensure that everything is carried out as per plan. Controlling is a continuous process which measures the progress of operations, (compares) verifies their conformity with the predetermined plan and takes corrective action if required.

Planning is the beginning of management process and controlling is the final stage.

Steps in controlling:

1. Establishing or setting standards of performance
2. Measuring actual performance of the enterprise
3. Comparing the actual performance with the standard set
4. Finding deviations, if any
5. Taking corrective action if required

6. Decision making

A decision is an act of choice where in a manager forms conclusion about what must be done under a given situation. A decision represents a course of behaviour chosen from a number of possible alternatives.

Decision making is a very important function right from the stage of planning till the stage of actual production and marketing. The decision makes every function a grand success or a miserable failure. Sometimes even a wise decision may result in failure due to unforeseen reasons which are beyond the scope of human control. Hence before making any decision one should be thorough with the past, think of present and plan for the future with the available past information. Sometimes to take long term decisions the workers are also involved in the process of decision making. i.e., with regard to the working conditions, change in layout, methods of production, quality of raw material, etc.

Steps in Decision making:

1. Defining the problem.
2. Analysing the problem
3. Developing alternative solutions
4. Deciding upon the best solution
5. Converting the decision into effective action
6. Implementing and verifying the decisions

Scientific Approach to Decision Making:

The scientific approach is a formalized reasoning process. It consists of the following steps:

Step 1: The problem for analysis is **defined** and the conditions for observation are determined.

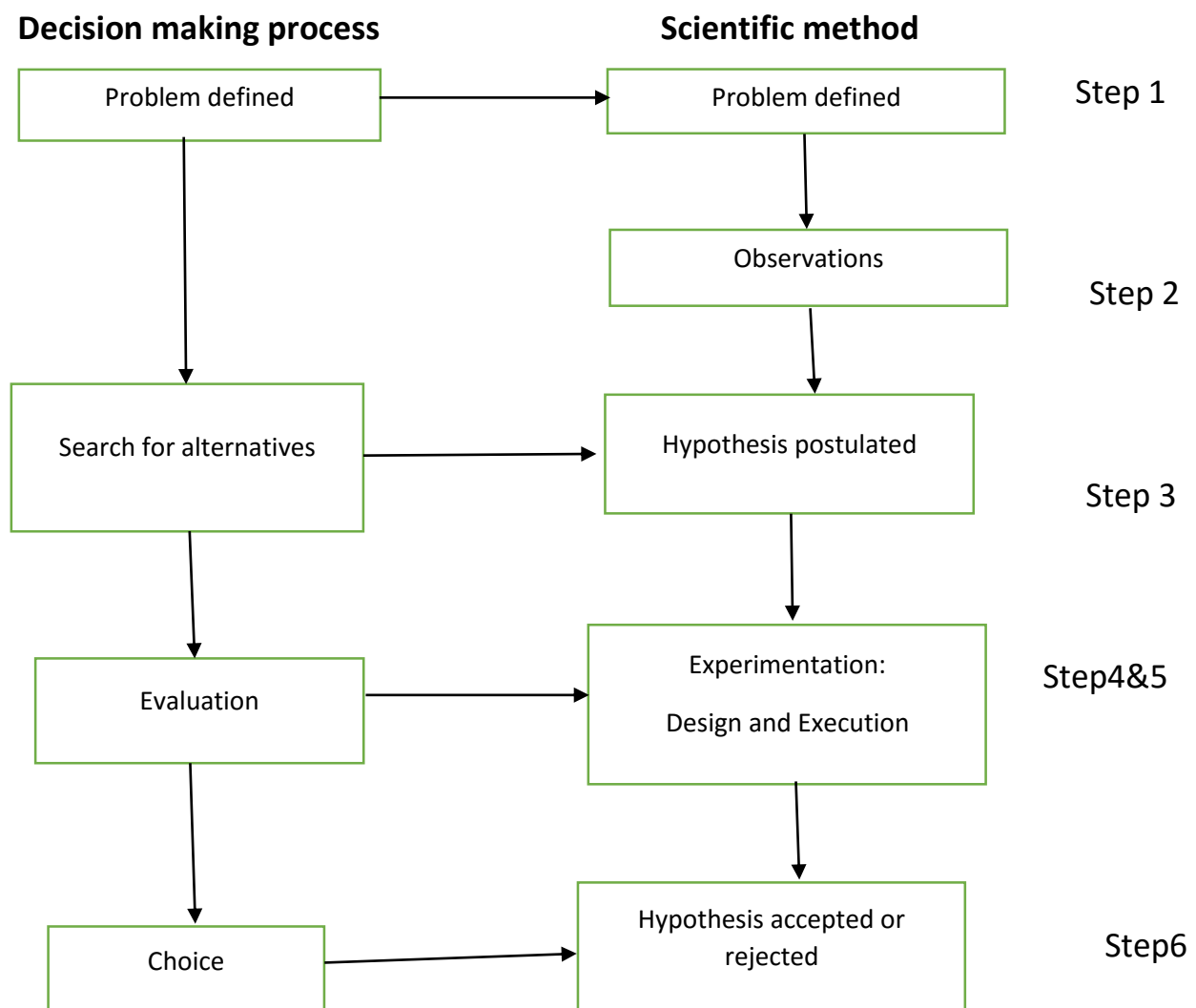
Step 2: Observations are made under different conditions to determine the behaviour of the system containing the problem.

Step 3: Based on the observations, a **hypothesis** that describes how the factors involved are thought to interact or what is the best solution to the problem, is conceived.

Step 4: To test the hypothesis, an **experiment** is designed.

Step 5: The experiment is executed and **measurements** are obtained and recorded

Step 6: The **results** of the experiment are analysed and the hypothesis is either accepted or rejected.



OWNERSHIPS

Types of Ownerships are:

1. Individual or sole proprietorship
2. Partnership Organization
3. Joint stock company
 - a) Private Ltd. Company.
 - b) Public Ltd. Company.
4. Cooperative Society.

1. Individual ownership or Sole proprietorship

The business which owned and controlled by a single person with or without hired labour is called Sole Proprietorship. The person who runs sole proprietorship is called sole trader.

Advantages:

1. Easy for formation as it does not require to complete any legal formality.
2. Owner is free to make any decisions.
3. Simplicity in the organization and expenses in starting are minimum.
4. The owner can meet the customer's taste directly.
5. The secrecy can be maintained (material, process, method etc.)
6. It is more flexible and product variety can be changed.
7. The efforts and rewards are directly related.
8. Minimum legal restrictions.
9. Since the supervisor is the proprietor himself, the overhead expenses are very less and products can be obtained cheaply.

Disadvantages:

1. The owner cannot be master of all techniques like management, sales, engineering, process etc. Hence work suffers.
2. If the business fails, creditors can take the personnel property as well as business property of the owner to settle their claim. So this involves unlimited liability for debts and losses.
3. The capital is limited, therefore modern firms with sophisticated machines cannot run with this type of organization.
4. The firm may stop with the death of the proprietor.

Application:

1. When the business can be taken care by one single person in all functions of management.
2. Enterprises requiring small capital.
3. For business which don't involve high risk or failure.

Example: Starting a printing press, auto repair shop, small fabrication/ engineering industry, wood working shop etc.

2. Partnership

It is defined as the relationship between persons who have agreed to share the capital and profit of business carried on by all or any of them acting for all. The persons participating in the business are called individually as partners and collectively the business called firm. The minimum number of persons to start partnership is 2 and maximum is 10 in case of banking business and 20 in case of other business.

Types of partners:

1. **Active Partners:** Who take active part in the management of the business enterprises
2. **Sleeping Partners:** Who do not take any active part in the business process, but invest their money.
3. **Nominal Partners:** Who do not invest money and do not take part in the management, but they lend their reputed name for the company's product/service.
4. **Secret Partners:** Whose name do not appear anywhere, but they took part into the management secretly.
5. **Minor Partner:** Who has not attained the age of 18 years.
6. **General Partners:** All the partners in the organization are general partners.

Advantages:

1. Large capital can be collected than that of the sole trade.
2. The firm possess much better talent / skills from different partners.
3. There is a definite legal status.
4. Partnership can borrow money easily from various financial institution.
5. Incentive for success is high.
6. Partnership associates tax advantages with it.
7. Not subjected to strict government supervision.
8. For all losses there are more than one person to share

Disadvantages:

1. Mistake of one partner may cause a big loss to all the partners.
2. Due to unlimited liability, risk involved is more.
3. Chances of misunderstanding among partners which affect adversely on the efficiency and expansion of business.
4. It is unsuitable for modern industries because they require huge capital.
5. Partnership comes to an end if a partner dies.
6. Investors and lenders hesitate to provide money because of lack of stability of a partnership firm.

Application:

1. Medium engineering firms
2. Law firms
3. Cinema theatres
4. Educational institutions
5. Medical clinics
6. Retail traders etc.

Partnership deed:

To avoid any complication at a later stage, the constitution of the company may be written in an agreement form. This agreement is known as partnership-deed.

The partnership deed contains the following:

- Name of the firm.
- Name of the Business.
- Date of Starting Business.
- Money contributed by each partner.
- Allotment of management functions among partners.
- Salary if any, allowed to manage partners.
- Rate of Interests on capital invested, if any.
- The basis for the introduction of any new partner.

3. Joint stock company

A joint stock company is an association of several persons called shareholders who joined together for profit and agree to supply capital divided into shares that are transferable for carrying a specific business.

It comes into existence only after registration under Indian companies act or under special act of legislature.

The company has a continuous existence. It is not affected by the death, insanity or insolvency of any of its shareholders or directors.

Ordinarily, the members of the company are not held liable for its debts hence limited liability of members.

The shareholders are the owners of the company. As they are large in number, all of them can not directly take part in the management of the company. Hence, in a democratic way they elect few directors who manage the affairs of the company.

Since the shares are of small value, these shares are purchased by large number of people from every walk of life. And there is no maximum limit to membership in a public company, it can raise large amount of capital.

For the protection of the shareholders, creditors, and society in general, the government has provided statutory control over the affairs of companies.

Types of joint stock companies:

i. Private limited company.

Actually, a private joint stock company resembles much with partnership and has the advantage that big capital can be collected, than could be done so in partnership.

The capital is collected from the private partners (Maximum members 50); some of them are active while others being sleeping.

Transfer of shares is limited to members only and general public cannot be invited to subscribe the shares.

The company need not make the prospectus, accounts and other particulars open to public but it must get its accounts audited.

The members only are entitled to receive a copy of the balance sheet, and auditor's report.

The government does not interfere in the working of the company.

ii. Public limited company

In Public Limited Company, the capital is collected from the public by small shares of value of Rs.5, 10, 20, 50 etc.

The number of shareholders should not be less than seven, but there is no upper limit.

Shares are transferable in part or full without requiring any prior approval.

The affairs of the company are managed by an elected body known as 'Board of Director'.

The directors of the company (Maximum 7 directors) are subject to rotation.

A public company has to issue a prospectus, to public and it should send the financial statements to all members and to the registrar.

It must get its accounts audited every year by registered auditors.

It can start only after receiving the 'Certificate to Commence Business'.

Advantages:

1. The liability being limited, the shareholders bear no risk and more and more persons are encouraged to invest capital. So huge amount of capital can be collected to run modern industries.
2. It has great potentialities for expansion.
3. Shares are transferable.
4. Specialist services can be utilized.
5. Risk or loss is divided among many shareholders.
6. It can bear big salaries and thus better administration.
7. Not affected by the death or retirement of the shareholders.

Disadvantages:

1. There are sufficient scope for the management for their personnel profits, because they know the financial position of the company, therefore they can purchase or sell the shares accordingly.
2. A good deal of legal formalities is required for the formation of joint stock company.
3. The team spirit with which partnership works, is lacking in a joint stock company.
4. It is difficult to preserve secret in these companies.
5. High paid managers cannot have high interest in the company as the proprietor can have.
6. People can commit frauds with the company.
7. Company is managed by big shareholders only.

Applications:

Fertilizer factories, Steel mills and heavy duty engineering concerns, etc.

Difference between Pvt.Ltd Company and Public Ltd Company:

	Private limited company.	Public limited company.
a)Numbers of members:	There should be minimum of 2 and maximum of 50 members.	There should be minimum 7 members and there is no upper limit to its membership.
b)Commencement of business;	A private company can commence its business soon after getting the certificate of incorporation.	A public company cannot commence business unless it obtains the second certificate, i.e. the Certificate of commencement of Business.
c)Issue of prospectus	A private company cannot issue a prospectus when inviting the public to subscribe to its shares and debentures, It can collect the share only from its limited number of members in advance.	A public company depends actively upon public subscriptions and must issue a prospectus.
d)Transfer of shares	In private company the shares are not transferable.	Whereas the shares of a public company are freely transferable.
e) Number of directors	A private company there is no limit on directorship.	Whereas a public company is limited to 20 directors.
f) Statutory meeting	A private company is not required to hold statutory meeting	Public company has to hold its statutory meeting within a specific period after the commencement of business.

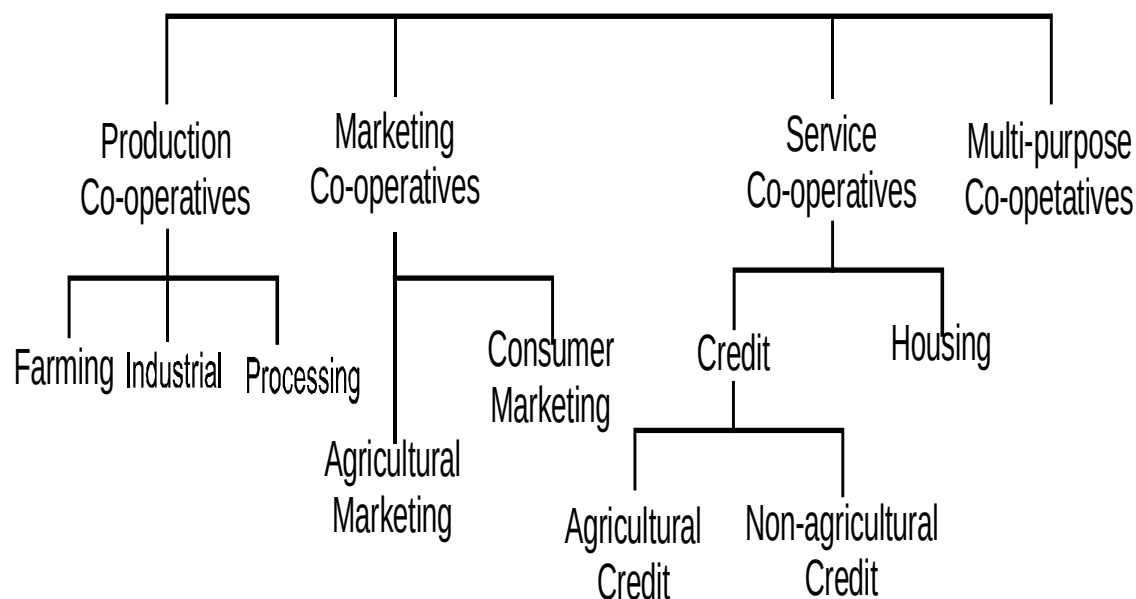
4. Cooperative societies

- To achieve common economic objectives, a set of people may associate together and form a cooperative society.
- They may be agriculturists, artisans, consumers, Laborers etc.
- All these societies are registered and regulated by the cooperative society's act of the concerned state.
- The main aim of a cooperative organization is self-help and mutual help and not for making profit.
- The slogan of cooperation is "one for all and all for one".

Principles of cooperative societies:

1. Voluntary association.
2. Democratic management.
3. Not profit motive.
4. Self-help and mutual help.
5. Open door policy.
6. Distribution of Surplus.

Classifications of cooperative societies:



How to start a cooperative society?

- An application should be submitted to the registrar of cooperative societies of the state. The important condition for registration of the society is, there must be at least 10 members who are competent to contract.
- The application should state clearly the name, place, the objectives, capital structure, source of funds of the society etc.
- Byelaws is to be drafted for the society. These byelaws are the rules and regulations in accordance with which the society shall function, and they are somewhat like the memorandum and articles of association of a company.
- Along with the application, two copies of byelaws and the prescribed registration fees are submitted to the registrar.
- If the registrar is satisfied, he will issue a certificate of registration after which the society can admit members, collect share capital and commence business.

Advantages:

1. Its formation is easy
2. It is service motive.
3. It benefits weaker sections, as daily necessities of life can be made available at lower cost
4. It is democratic form of ownership
5. Overhead expenses are reduced due to free services rendered by the members
6. It is subjected to state control and supervision
7. Middlemen's profit is eliminated
8. It is having limited liability.
9. It has open membership.

Disadvantages:

1. Finance being limited, specialist's services cannot be taken.
2. Conflicts may arise among the members on the issue of sharing responsibility and enjoying authorities.
3. Members who are in higher position may try to take personal advantages
4. Capital is limited and so it is limited to small business only.
5. Secrecy cannot be maintained
6. Lack of incentive to hard work.

ORGANIZATION STRUCTURE

Organization is defined as the machinery which establishes the coordination between administration and management.

Organization combines (organize) the various factors of production into the most effective coordination.

This machinery fixes the responsibility and authority to the workers and managerial people and makes them linked in a particular structure.

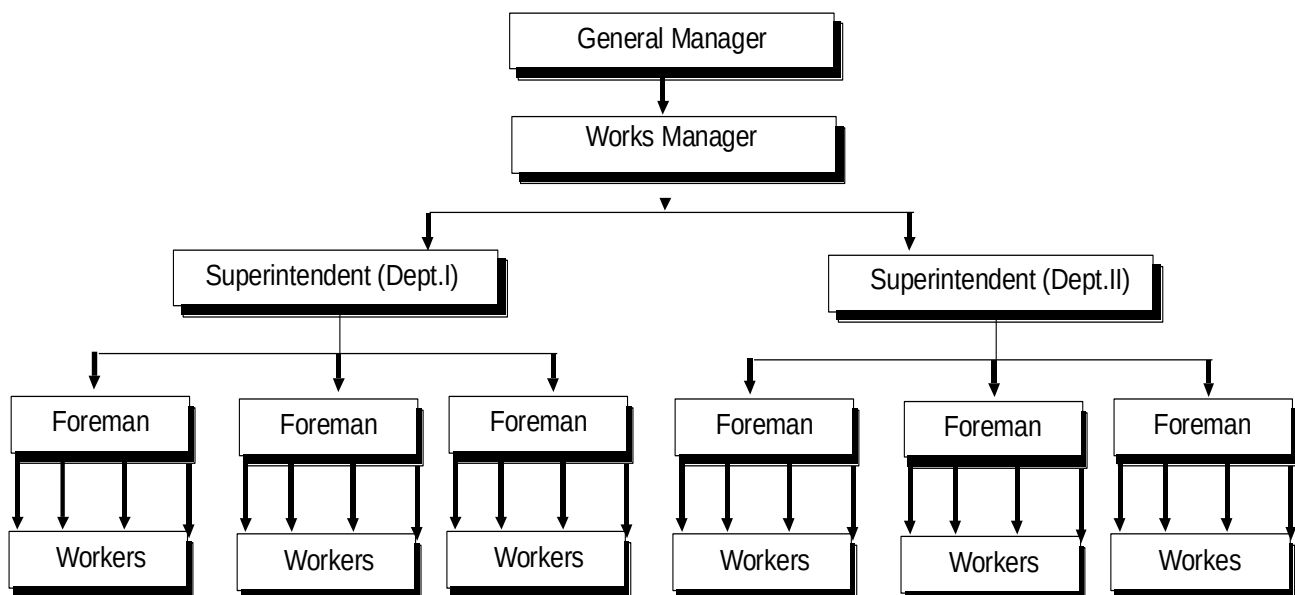
Organization build up a committed work force to achieve the target of the firm.

Different types of organisation:

1. Line or Military or Scalar organization.
2. Functional organization.
3. Line and Staff Organization.

1. Line or military or scalar organisation

It is the simplest form of organization structure. It based upon relative authority and responsibility rather than on the nature and kind of operation or activities. Line organisation is direct and people at different levels know to whom they are accountable. The immediate superior (or boss) gives orders to the subordinates, assigns duties, dismisses and takes disciplinary action against them.



Advantages:

1. It is simple and very easy to understand.
2. It is flexible.
3. It gives clear division of authority and responsibility.
4. It permits quick decisions and speedy actions.
5. Strong in discipline like in military.
6. Shifting of responsibility is not possible.
7. It is capable of developing the all-round executive at the higher levels of authority.
8. It gives increased efficiency and operations.

Disadvantages:

1. It neglects specialist role.
2. Overloads few key executives.
3. Chances of wastage, accident and labor turnover are more, because of insufficient knowledge of all the work by one man.
4. No means for rewarding good workers.
5. Department heads are overloaded with various routine works, hence no time to think for future expansion and planning.
6. As the orders flow from top to bottom chance of loss and distortion of information possible.

Applications:

1. It is suitable for continuous process industry, like sugar, paper and oil refinery and automatic plants.
2. It is also suitable for educational institutions and industries which are free from complexities.

2. Functional organisation

F.W. Taylor suggests functional organisation because it was difficult to find all-around expert persons qualified to work at middle management level in the line organisation.

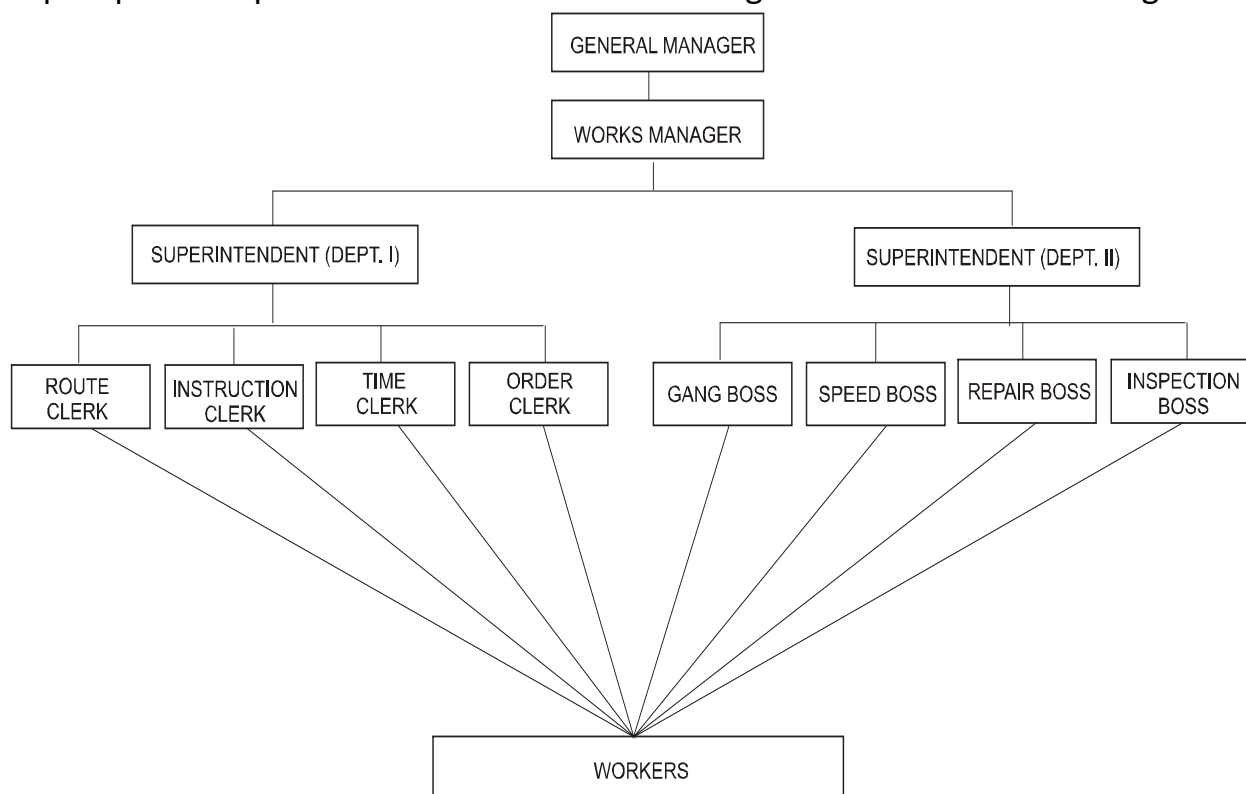


Fig 1.9 Functional organisation structure

Advantages:

1. It is based on expert knowledge.
2. Expert's guidance leads to better quality products and reduces wastage, accidents and man-machine hours.
3. It helps in mass production by standardization and specialization.
4. Unnecessary overloading of responsibilities will not be there, so top executives get time for planning for future expansion.
5. No special knowledge of workers is required as the instructions are supplied by experts.
6. Since foreman is responsible for one function, he can perform his duties in a better manner.
7. If any operation needs improvements, it can be improved even up to the last moment.
8. It helps mass production by standardization and specialization.

Disadvantages:

1. It is difficult to maintain discipline.
2. Functional experts may feel himself to be superior than the others, so that coordination of the effort of various experts is difficult.
3. Workers always remain confused about the authority and activity of each foreman.
4. Shifting of responsibility is possible.
5. Overlapping of responsibility is possible.
6. Failure of one expert will largely effect the production
7. As the workers are not given opportunity to make use of their ingenuity, initiative and drive their full capacity cannot be utilized

Application:

1. In practice a pure functional organisation is rarely found.
2. A combination of line and functional organisation can be seen, where responsibilities are divided into functional basis and keeping a line relationship.

3. Line and staff organisation

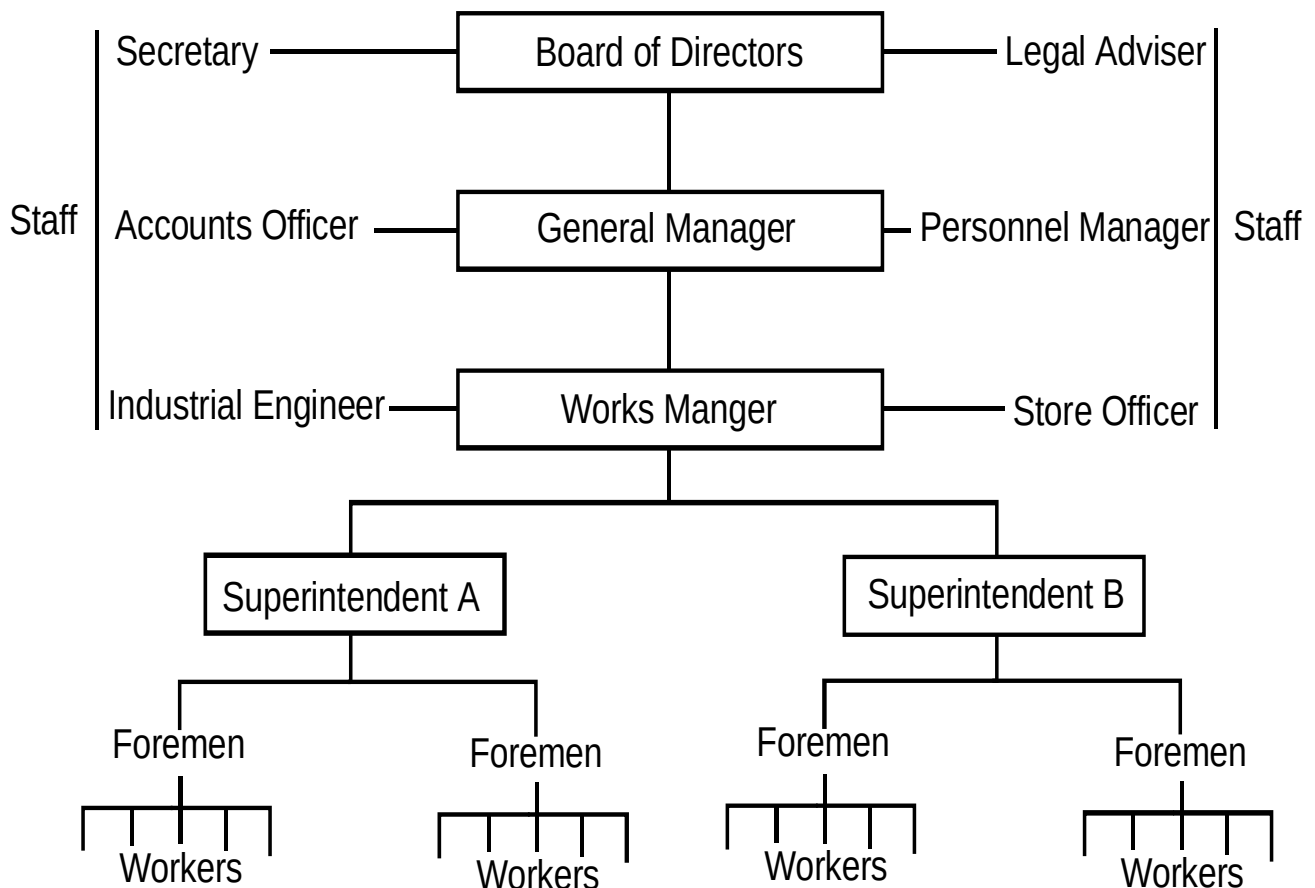
In large organisation operating on a big scale, managers cannot give careful attention to every part of the management. They are unable to think and plan.

They are busy with ordinary task of production and sales. Hence some staff is deputed to do the work of investigation, research, recording and advice to managers.

Thus staff brings specialization by assisting the line officers. The line maintains the discipline and stability.

Staff provides expert information and helps to improve the overall efficiency.

Thus the staff are the thinkers and the line are doers. Usually the staff reports to the executive and gives advice on the subject of his specialty.



Advantages:

1. It is a planned specialized system, so expert knowledge is available.
2. Line executives are relieved of some of their loads and are thus able to devote more attention towards production.

3. Less wastage of material, man and machine hours.
4. Quality of product is improved.
5. Discipline problem is solved because of line relationship.
6. There is no confusion as in the functional organisation.
7. It possess all the advantages of line and functional organisation.

Disadvantages:

1. Product cost will increase because of high salaried staff executives.
2. In case functions are not clear it may create confusion.
3. Friction and jealousies if developed between line and staff executives may cause harm to the enterprise.
4. Line executive if they start depending too much on staff executives may lose their initiative drive and ingenuity.
5. Sufficient expert knowledge and guidance is not available as compared with functional type organisation.

Applications:

Line and staff organisation is very common among medium and large enterprises.

Requirement of a good organisation

- A good organisation makes administration easy by dividing duties and responsibilities of individual worker according to his status in the factory.
- By assigning duties and allotting paper work to individuals according to suitability, education and experience. It promotes specialisation. This result in increased output and quality.
- It keeps coordination among the employees of different levels, which results in a healthy relations and harmonious and smooth atmosphere.
- It allows sufficient time for top management and administration
- To work for future expansion, planning and improvements.
- It increases margin of profit.
- It ensures discipline in all activities and prompt delivery.
- It clearly defines the authority, responsibility. It avoids confusion, duplication, wastage and inefficiencies.
- Organisation facilitates the flow of information and decision making from one level to another without delay and distortion.
- It promotes total involvement, commitment and mutual help.
- Span of control will be appropriate.

❖ LESSON 2. HUMAN RESOURCE MANAGEMENT(HRM)

Human resource management strives to use human resources effectively in order to exceed the target. In short managing man is the personnel management. Human Resource Management is also called as personnel management and it is also known as, Labor management, Manpower management, Personnel Administration etc.

Importance of HRM:

1. It discovers the talented experienced, qualified, competent workers to take up various jobs.
2. It ensures maximum productivity per worker by providing right man on the right job.
3. It helps in foreseeing manpower requirements for the present and for the near future.
4. It develops personnel for higher positions with greater challenges and responsibilities.
5. It enables best possible utilization of human resources resulting in reduction of labor cost per unit of production.
6. It gives workers, job satisfaction and essence of happiness due to proper placement.
7. It ensures attractive pay which provides mental satisfaction to workers.
8. It keeps the management informed of the present and future requirements due to transfers, promotions, labor turnover, or death etc.

Functions of Human Resource Management:

1. Managerial functions:

- Planning
- Organizing
- Directing
- Controlling

2. Operative functions:

- Procurement of personnel
- Development of personnel
- Compensation of personnel
- Employee's benefit schemes
- Maintaining good industrial relations
- Record keeping

- **Personnel planning and evaluation**
- **Personnel research and audit**

Major components of HRD:

- (i). Recruitment
- (ii). Selection
- (iii). Placement
- (iv). Induction and orientation
- (v). Performance appraisal
- (vi). Training
- (vii). Career planning and development
 - Transfer
 - Promotion
 - Demotion
- (viii). Compensation
 - Wages and salary administration
 - Incentives
 - Bonus
 - Fringe benefits
- (ix). Social and cultural programmes
- (x). Employee counselling
- (xi). Monetary rewards

MANPOWER PLANNING

The process by which management determines how the organization should move from its current manpower position to its desired manpower position. It deals with calculations of manpower requirements in future, considering various factors such as internal and external economies, social and political pressures, organization policy, availability and suitability of the manpower requirements, etc. it is a continuous dynamic process and manpower requirement should be done every year and revised periodically at fixed interval keeping in view the yearly manufacturing programme.

1. **Forecasting:** - Future manpower requirements are to be found out by considering the economy and developments in the industry. It may also consider future plans, diversifications, extensions, collaborations etc.
2. **Inventorying:** - Present manpower resources available are to be considered and analyzed for effective utilization.

3. **Investigating:** - Investigate the job requirements and job descriptions and the qualities required for each and every job.
4. **Anticipating:** - Manpower problems are analyzed comparing the present resources into future.
5. **Planning:** -The necessary programs of recruitment, selection, training, development, motivation and compensation so that the future manpower requirements are met.
6. **Selecting:** - Selecting the appropriate sources of recruitment.

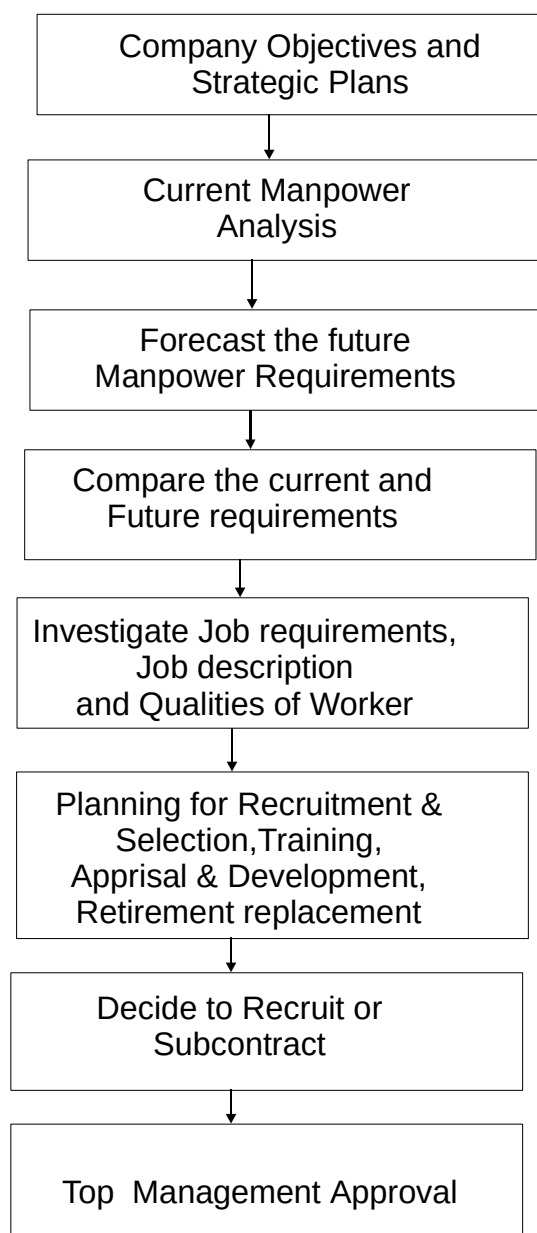
Objectives of manpower planning:

1. To ensure optimum use of human resources currently employed.
2. To forecast future skill requirements.
3. It should be reviewed periodically, so that modification or alternations if any can be incorporated.
4. It provides control measures to ensure that necessary resources are available as and when required.
5. To determine the recruitment level.
6. It should be need based.
7. It should be efficient and effective in nature.
8. To determine optimum training levels.
9. To cost the manpower in projects.
10. To assist productivity bargaining.
11. To link manpower planning with Organization planning.
12. To decide whether certain activities need to be subcontracted.

Factors affecting manpower planning:

1. Working hours
2. Number of shifts
3. Nature of production
4. Product mix
5. Performance rate
6. Hours lost

Process of Manpower planning:

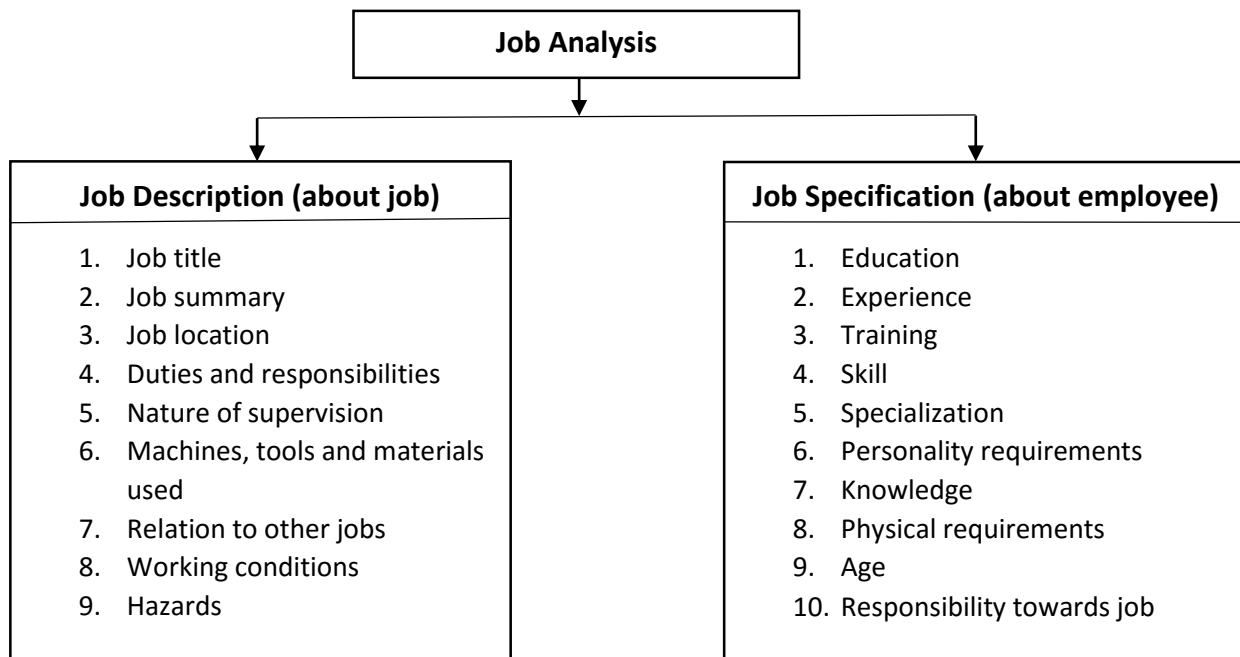


JOB ANALYSIS

Job analysis is the breaking up of a job into basic elements or operations, and studying in detail each of the operations to know the nature and characteristics of the job. Further it is the study of the skill, duties, responsibilities, involved in the job and the qualification required in the worker for the better performance of the job. Thus, it is the study of the job and the employee.

The information required for the job analysis can be collected by following methods:

1. Observation method
2. Individual interview method
3. Group interview method
4. Structured questionnaire method
5. Technical conference method
6. Dairy method



JOB EVALUATION

In every Organisation there are different types of jobs having different characteristic. Depending upon the job characteristics factors like qualification, skill, experience physical ability, aptitude may vary. Therefore, the various jobs in an industry are to be analyzed and classified into different grades based on their requirements and varying factors. This process is known as job evaluation which helps in rewarding the job according to its real value. It is the measuring of monetary worth of a job and determining the wages for the job. So it shows the relative value of different jobs in the Organisation.

What is job evaluation and What it is not:

- Job evaluation is concerned with **job content or demands** of the job and *not* the '**Value**' of the job in the Organisation
- Job evaluation **rate the job** not the **man** who performs it.

- Job evaluation, despite any qualification, is a **disciplined judgment** about the hierarchical positioning of jobs.

Objectives of Job evaluation:

1. To decide relative values of different jobs
2. To determine a suitable wage structure.
3. To improve employer employee relations
4. To reduce labor turnover and improve job satisfaction.
5. To minimize labor unrest and to create healthy work situations

Methods of job evaluation

1. Non-Quantitative type job evaluation

There are two types of non-quantitative type evaluation as follows:

a) Ranking system

b) Rating or Grading or Classification system.

a) Ranking System

This is the simplest method. The job to be evaluated are ranked according to the value of the work as judged by the rankers. The ranking may be done by a trained analyst or by a committee of competent men. In ranking a job, the following factors may be considered.

1. Volume of work.
2. Difficulty of work.
3. Working conditions
4. Responsibility involved
5. Monotony of work.
6. Working conditions.

b) Rating or Grading or Classification System

A brief description of such a classification in an office

- I. Class I - Executives: Further classification under this category may be Office Manager, Deputy office manager, Office superintendent, Departmental supervisor, etc.
- II. Class II - Skilled workers: Under this category may come the Purchasing assistant, Cashier, Receipts clerk, etc.
- III. Class III - Semiskilled workers: Under this category may come Steno typists, Machine-operators, Switchboard operator etc.
- IV. Class IV - Semiskilled workers: This category comprises Drafters, File clerks, Office boys, etc.

2. Quantitative methods of Job evaluation

a) Factor Comparison b) Point or Manual System.

a) Factor Comparison

Under this method, instead of ranking complete jobs, each job is ranked according to a series of factors.

These factors include mental effort, physical effort, skill needed, responsibility, supervisory responsibility, working conditions and other such factors (for instance, know-how, problem solving abilities, accountability, etc.).

Pay will be assigned in this method by comparing the weights of the factors required for each job, i.e., the present wages paid for key jobs may be divided among the factors weighted by importance (the most important factor, for instance, mental effort, receives the highest weight).

In other words, wages are assigned to the job in comparison to its ranking on each job factor.

The steps involved in factor comparison method:

- I. Select key jobs (say 15 to 20), representing wage/salary levels across the Organisation. The selected jobs must represent as many departments as possible.
- II. Find the factors in terms of which the jobs are evaluated (such as skill, mental effort, responsibility, physical effort, working conditions, etc.).
- III. Rank the selected jobs under each factor (by each and every member of the job evaluation committee) independently.
- IV. Assign money value to each level of each factor (example: consider problem solving is one of the factor, what level of problem solving is required {basic, intermediate or advance}) and determine the wage rates for each key job.
- V. The wage rate for a job is apportioned along the identified factors.
- VI. All other jobs are compared with the list of key jobs and wage rates are determined

b) Point or Manual System

In the system, the value of the job is expressed in terms points. This method is very popular, systematic and accurate. This method consists of

1. Divide the job into number of factors.
2. Award certain points for each job.
3. Arriving at total points for each job.

4. Determining a suitable wage rate proportional to the total points.
5. Classify the total points into grades as given below.

Points evaluation for a set of jobs:

Job	Mental	Skills	Physical	Responsibility	Working condition	Total Points
Machinist	100	80	20	50	50	300
Fitter	80	80	40	40	20	280
Assembler	80	80	30	40	50	280
Welder	100	90	50	50	50	340
Attender	20	10	80	40	30	180

For the above hypothetical example, the total points vary from 180 to 300. This range may be divided into four grades:

Grade - I 150 - 199 points

Grade - II 200 - 249 Points

Grade - III 250 - 299 Points

Grade - IV 300 - 349 Points

Merit Rating or performance appraisal

- As explained earlier the job evaluation measures relative worth of jobs without considering the merits of the person doing the job.
- Here merit rating assesses the merit of the person doing the job.
- It measures the loyalty, integrity, intelligence, leadership qualities, dependability, character etc. i.e., it measures the extent to which an employee meets job requirements.
- So job evaluation and merit rating are two complementary aspects of a sound personnel policy.
- Merit rating **gives** a basis for determining wage increment, promotion, training, transfer, special incentives etc. Merit rating is commonly known as employee rating, employee appraisal, performance appraisal or staff reporting.

- Merit rating is done for the matters of -
- | | |
|--------------------|------------------------|
| 1. Wage increments | 2. Promotions |
| 3. Transfers | 4. Special assignments |
| 5. Training | 6. Discharge |

Method of merit rating:

Rating scale method is generally used for merit rating. The steps involved in this method are -

1. Merit factors to rate the employees are decided. The different factors are: -

- | | |
|------------------------|---------------------|
| 1. Standard output | 2. Character |
| 3. Quantity of output | 4. Integrity |
| 5. Intelligence | 6. Loyalty |
| 7. Job knowledge | 8. Cooperation |
| 9. Initiative | 10. Reliability |
| 11. Creative ability | 12. Dependability |
| 13. Education | 14. Leadership |
| 15. Experience | 16. Judgment |
| 17. Organizing ability | 18. Enthusiasm etc. |

Six to ten factors, depending on the nature of job, may be used for rating an employee.

2. Each factor is divided into different grades like excellent, very good, good, fair, poor etc., For each grade certain points or marks are fixed.
3. Merit factors are selected for an employee to be rated. Marks are awarded by the raters, for each factor.
4. Total marks obtained is the rating of that employee.

Other two types of Merit Rating Methods are:

- 1. Checklist Method**
- 2. Employee Comparison Method**

TRAINING

After employee is selected, placed, and introduced he or she must be provided with training facilities. "Training is the act of increasing the knowledge and skill of an employee for doing a particular job".

Importance of Training:

The training of Employees is essential in order to:

1. Ensure an adequate supply of properly trained employees at all levels of industry.
2. Improve productivity and quality.
3. Attain precision and clarity in the transition of business.
4. To reduce wastage, accidents, fatigue, labor turnover, absenteeism and overtime.
5. To boost employee morale, cooperation and good relations.
6. To inculcate a broad understanding of relevant science and technology.
7. To teach standardized work methods.
8. To promote team work.
9. To inculcate good work habits.
10. To improve the performance of each employee to the highest level.

Advantages of Training:

1. To increase the efficiency of workers/supervisors.
2. To reduce wastage of materials, machine and man hour.
3. To increase productivity and reduces production cost.
4. Reduced supervision and improved product quality
5. It gives job satisfaction.
6. It reduces labor turnover and chances of accidents.
7. Less fatigue to the workers.
8. Increased organizational stability and flexibility.
9. Specialization and standardization is easy.
10. It can boost the morale, cooperation and good relation.
11. It helps to build team spirit.

Methods of Training:

1. On-the-job method

- a) **Job rotation:** This type of training involves the movement of the trainee from one job to another.
- b) **Coaching:** The trainee is placed under a particular supervisor functions as a coach in training the individual.
- c) **Job instruction or training through step by step:** This method is also known as Training through Step-by-Step. In this method trainer explains the trainee the way of doing the jobs, job knowledge and skills and allows him to do the jobs.
- d) **Committee assignments:** A group of trainees are given and asked to solve an actual organizational problem. The trainees solve the problem jointly. It develops a team work.

2. Off-the-job method

- a) **Vestibule training:** Actual work conditions are simulated in a class room. Material, files and equipment those are used in actual job performance are also used in training.
- b) **Lecture methods:** The instructor organize the material and gives it to a group of trainees in the form of a talk.
- c) **Role playing:** It is defined as a method of human interaction that involves realistic behaviour in imaginary situations.
- d) **Conference or discussion:** It is a method in training the clerical, professional and supervisory personnel. This method involves a group of people who pose ideas, examine and share facts, ideas and data, test assumptions, and draw conclusions, all of which contribute to the improvement of job performance.
- e) **Programmed instruction:** The subject matter to be learned is presented in a series of carefully planned sequential units. These units are arranged from simple to more complex levels of instruction. The trainee goes through these units by answering questions or filling the blanks. This method is expensive and time consuming.

LABOUR TURNOVER

This may be defined as the ratio of number of workers who have left their job on their accord to the average number of workers employed in a factory during a given time.

$$\text{Labor Turnover} = \frac{\text{No.of workers left the organization during the period}}{\text{Average no.of workers in the organization during the period}} \times 100$$

Causes of Labor turnover:

1. Fluctuation in volume of work
2. Long hours of work and poor wages
3. Bad working conditions
4. Unsatisfactory procedure of discharging a worker
5. Failure to protect the workers against accidents and sickness
6. Inefficient management
7. In subordination
8. Bad system of recruitment
9. Other reasons like-better job elsewhere, poor health, conditions at home, retirement, death, discharge or dismissals.

Methods to reduce labor turnover:

1. Try to minimize fluctuation of work load
2. Provide better wages and service conditions
3. Reasonable hours of work
4. Workmen's compensation act must be followed
5. Scientific method of selection and training of workers
6. Provide safer working areas which reduces accidents
7. Complaints of the workers may be attended at the grass root level
8. Motivate workers both financially and non-financially
9. Be impartial in the matter of transfer and promotions.

❖ LESSON 3. WAGES AND INCENTIVES

The objectives of a good wage payment system reveal its importance as follows:

- It enables the employee to earn a good and reasonable salary or wage
- It pays equitable sums to different individuals, avoiding anomalies
- It should be understandable and acceptable to the employee
- It rewards and encourage high quality work
- It encourages employees to accept the changes in methods of working
- It encourages employees to use their initiative and discretion.
- It encourages employees to develop better methods of working.
- It rewards and encourage high level of outputs.
- It motivates a team spirit and cooperation.
- It improves employer-employee relations.
- It gives job satisfaction and job enrichment.
- It promotes peace and harmony in the Organisation.
- A good wage system will make the administration easier.
- It improves overall efficiency of the Organisation.

TYPES OF WAGES:

1. **Nominal Wages:** It is the amount of money paid to a worker in cash for the effort put in by him in an industry and no other advantage to the worker is made. This is also called money wages
2. **Real Wages:** Real wages includes the amount needed to meet the necessities, comforts, luxuries and cash payments which a worker can get in return of his effort and work.
For example, uniforms, essential commodities, housing with free water and electric supplies, conveyance and other such facilities are generally provided by the factory, in addition to the money in cash.
All these amount as a whole is considered as Real wages.
3. **Living Wages:** The rates of the wage are in such a way that they can meet some of the requirement of the family, like education, food, clothes and some insurance against the more important misfortunes along with prime necessities of life are called “Living Wages”.
4. **Fair Wages :-**It is actually the wage which must be fair for the work of a worker and should provide him with other necessities of life in addition to food for his family. The rates for the fair wages ranges between the minimum wage and living wage but between these two wages.

5. **Minimum Wages:** India government enacted Minimum Wage Act in 1948. According to this act wages have been fixed for different parts of the country and this has forced employers to give not less than this fixed minimum wages to any of the workers.

The main objectives of this acts are:

- a) To protect the low wage earners of the society whose conditions are materially unsatisfactory.
- b) To prevent exploitation of workers and to secure a wage according to the value of work done.
- c) To promote peace in industry.
- d) To improve normal standard of living.

Requirements of a Good - Wage Payment System:

1. It should provide rewards for good performance and incentives for further improvements in performance.
2. It should ensure equality in pay for similar jobs.
3. It must have the free consent of the employees.
4. It would be well planned and guarantee a minimum wage.
5. It should be simple, easy to understand, operate and control.
6. It must not involve heavy clerical work.
7. It should create appropriate differential between different level of jobs in accordance with their relative value.
8. It should operate flexibly enough to accommodate organizational changes and relations in the relative market rates for different skills.
9. It would be cost - effective in the sense that the benefits of the system are obtained without undue expense.
10. It should encourage suitable staff to remain with the Organisation.
11. It should take care general increase in the cost of living or in market rates by a proportionate increase in the minimum and maximum wage levels.
12. There should be incremental systems which will indicate the rates at which individuals can progress according to merit or experience.
13. There must be guidelines for determining merit increments and planning wage progression.
14. It should have clear salary review guidelines, the limit to which the pay of each department can increase.
15. It should aim at increasing production without adversely affecting quality.
16. It should reduce wastage of material, man hours, and machine hour.
17. A worker should get incentive both for quantity and quality.

18. Standardization should be the basis for all incentive scheme, Work method, materials, work place, working conditions etc. all should be standardized.
19. Incentives, bonus etc. should be payable along with the wages and not put off for future.
20. No limit should be put on a worker's incentive earnings.
21. The formula for calculating bonus and the conditions under which it is paid should be clearly defined.

INCENTIVE

Incentive may be defined as something that encourages a worker to put in more productive efforts voluntarily, or incentive is an inducement or reward which is given to a worker for his efficiency and hard work.

Incentives are in addition to the normal hourly rate and are in some proportion to the worker's contribution towards production.

There are mainly three types of incentives as follows:

1. Financial incentives
2. Non-Financial incentives,
3. Semi-Financial incentives

1. Financial Incentives

In addition to the extra payment for the immediate work contribution, workers may receive additional financial benefits in the form of:

- | | | |
|-----------|---------------|-----------------------|
| (i) Bonus | (ii). Premium | (iii). Profit sharing |
|-----------|---------------|-----------------------|

Methods of financial incentive payment:

1. Straight piece rate system
2. Straight piece rate with a guaranteed wage
3. Differential piece rate system
4. Halsey plan
5. Rowan plan
6. Gantt plan and bonus system
7. Bedaux plan
8. Emerson's efficiency plan
9. Group plan

2. Non-Financial incentives

In order to create interest in a worker for greater and better output, non-financial incentives must also be enforced and workers will also enjoy richer and fuller life. Non-financial incentives include:

1. Earning higher status.
2. Greater responsibility.
3. Chance of participation in decision making.
4. Token rewards like mementos.
5. Recognition in group.
6. Job satisfaction and security.
7. Better and healthy working conditions and surroundings.
8. Opportunity for quick promotion.
9. Training and future study opportunities.
10. Helpful and cooperative management

3. Semi - Financial Incentives

These are combination of financial and non-financial incentives. They include:

1. Provision of canteen at a subsidized rate, recreational and medical facilities to the workers and subsidized educational facilities for their children.
2. Cooperative societies for workers where subsidized consumable items will be obtained.
3. Conveyance facilities to employees at a concession rate.
4. Pension and other benefits etc.

FINANCIAL INCENTIVE PLANS FOR DIRECT WORKER:

1. Straight Piece Rate System

- Under this, a fixed rate of wage is paid for each piece or unit produced.
- **Earnings of a worker =**
No. of Pieces or Pieces or units produced × Rate per piece
- Example: If a worker fabricate 10 steel chair per day and for each chair the wage rate is Rs.10. Then he earns at the rate of Rs.100 per day (8 hours).

Advantages:

1. The method is simple, easy to understand and operate.
2. Workers are paid on their merits.
3. The administrative works are reduced.

4. It helps in boosting production and thus reduced the overhead expenses per unit of production.
5. Due to increased production profit is increased and selling price reduced.
6. The employer is enable to know his labor cost easily.
7. The relationship of the employer and employee improves and no dispute takes place for wages, as the workers get satisfactory reward for their work.

Disadvantages:

1. A worker is not guaranteed minimum wage
2. It is difficult to fix accurate and satisfactory piece rate.
3. With a high speed of work to produce more (and thus to earn more) the worker may not give proper attention towards the following:
 - a) Maintaining required quality.
 - b) Effective utilization of materials, equipment and tools etc.
 - c) His health
4. When the worker earns more, the employer may tries to reduce piece rate, which may cause labor problems.
5. It may cause increased accidents.
6. This caused displacement of labor as with increase in production, the number of workers engaged will be reduced hence no job security.
7. It cause over production and may result loses.
8. The entire benefit to the extra wages goes to the worker and no direct benefit to employer.

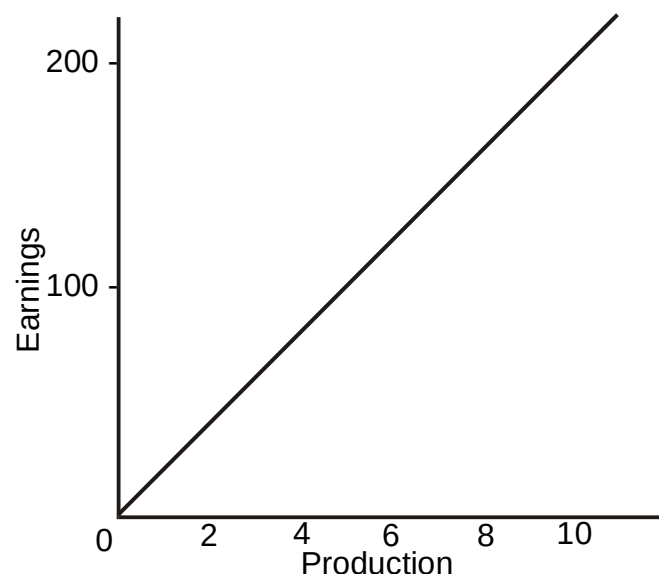


Figure: - Piece rate system

2. Straight piece rate with guaranteed minimum wage

This method is an improvement over the straight piece rate system as it guarantees a minimum hourly or daily base wage. Suppose that the standard output set by the management is 10 pieces per day. If a worker produces less than this amount he still gets the minimum wage and if another worker exceeds this standard, he is given a wage in direct proportion to the number of pieces produced by him at straight piece rate.

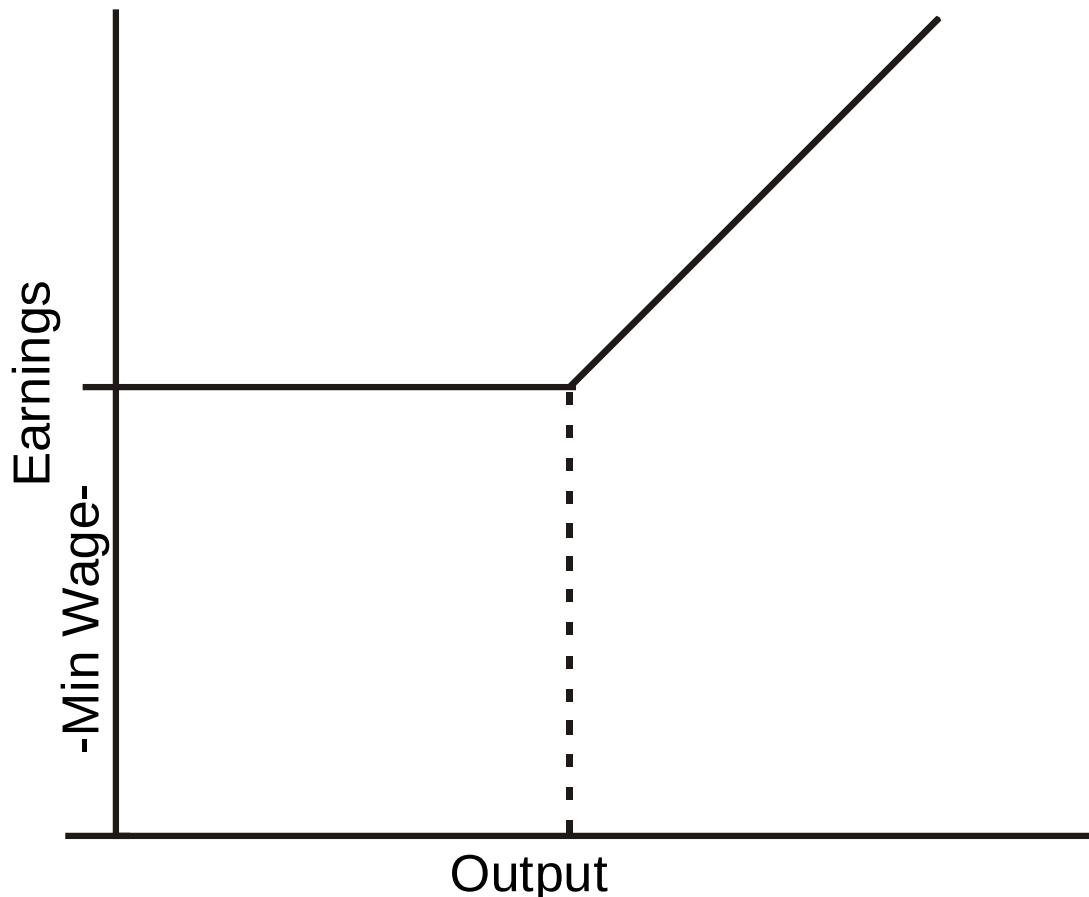


Figure: - Straight piece rate with guaranteed minimum wage

3. Differential piece rate system

This system suggested by F.W. Taylor, in which a standard time is fixed for the performance of a piece work and those who complete the job within standard time or produce the job earlier are paid at higher rates and those who do not complete the job in standard time are paid at lower rates.

This system, therefore gives an encouragement to active workers but punishes lazy workers. This system has the disadvantage that fresh workers, who are unable to reach output standard could earn very little and hardly service.

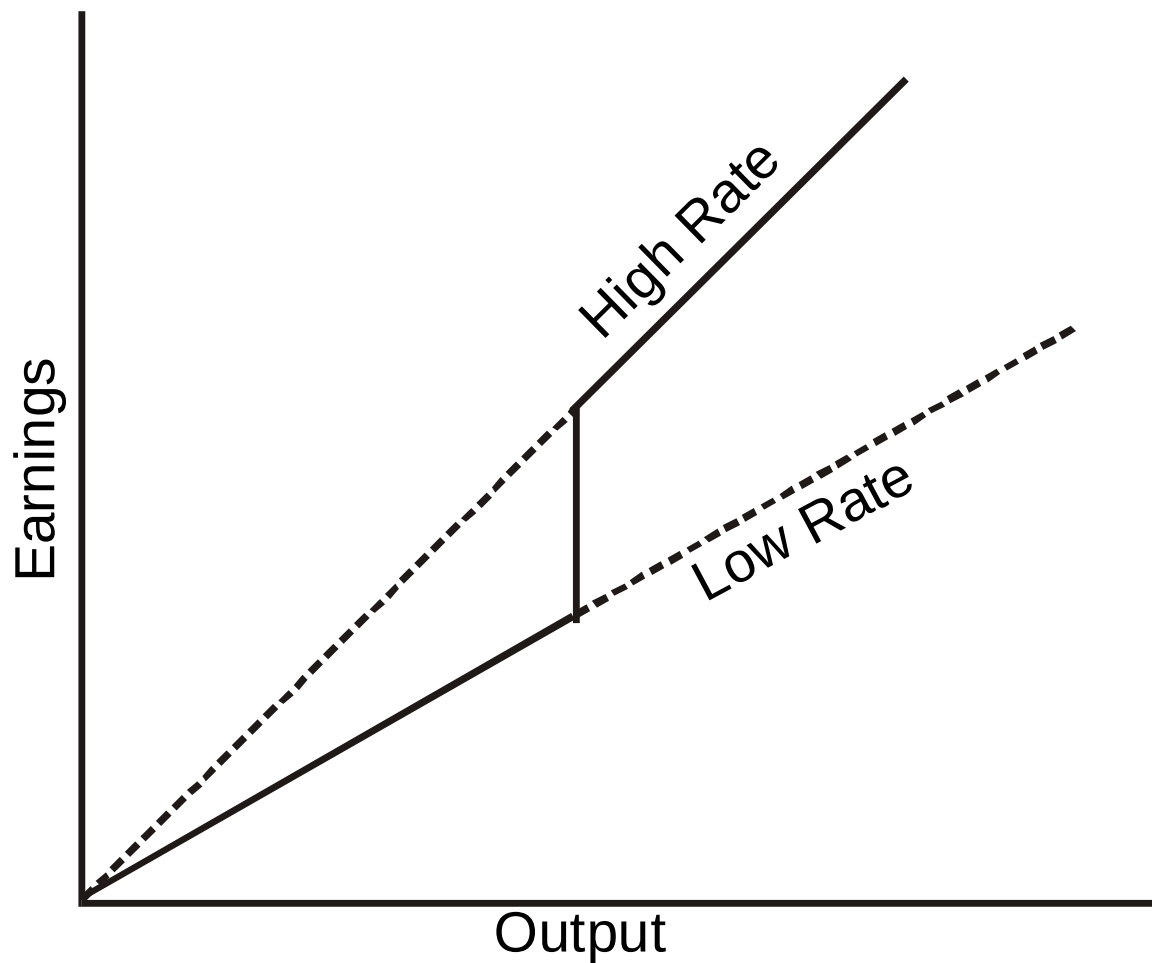


Figure: - Differential piece rate system

4. Halsey Plan

- In this an hourly rate or daily rate is guaranteed.
- Standard time is fixed for each job.
- The worker gets the agreed rate per hour for the time spend.
- An additional bonus or incentive is given to worker who completes the work earlier than standard time.
- The rate of bonus will be $33\frac{1}{3}$ % of time saved or 50% of time saved.

Wage of the worker = Normal earnings + Incentive

$$= RT + \left(\frac{P}{100} \right) \times (S - T) \times R$$

or a 50-50 Halsey plan

$$\text{Wage} = RT + \left(\frac{50}{100} \right) \times (S - T) \times R$$

$$= RT + \left(\frac{S - T}{2} \right) \times R$$

Where R = Hourly wage rate

S = Standard time

T = Actual time taken to complete a job

Note that 50% Incentive goes to the management.

For a $33\frac{1}{3}\%$ i.e., $\frac{100}{3}\%$ Halsey plan

$$\text{Wage} = RT + \frac{100 \times (S - T) \times R}{3 \times 100}$$

$$= RT + \frac{(S - T) \times R}{3}$$

Advantages:

1. It guarantees minimum wage.
2. It is simple to understand and operate.
3. It does not consume time on expensive time studies.
4. Management also share a percentage of bonus.

Disadvantages:

1. Workers do not like that management should share the bonus on the time saved solely because of their efforts.
2. Output standards or standard time is based upon past production records (and not in time study) may not be accurate and fair or just to all workers

5. Rowan Plan

1. This system also guarantees an hourly rate or daily rate.
2. Standard time is fixed based upon past production record similar to Halsey plan.
3. The worker gets an agreed rate per hour for the time spend.
4. An additional incentive is given = $\frac{(S-T)}{S}$ rather on $(S-T)$ of Halsey.

In Rowan Plan

Wages of worker = Normal Earnings + Incentive

$$W = RT + \frac{(S-T) \times RT}{S}$$

Where, R = Hourly Rate,

T = Time taken to complete the job,

S = Standard time or allowed time.

Advantages:

1. It can be employed when output standard is not accurate.
2. It provides a guaranteed minimum wage.
3. Inferior and fresh workers are not penalized.
4. Management also shares the bonus.

Disadvantages:

1. The worker do not like this plan because they do not get the full benefit of time saved.
2. To some extend it does not demark an efficient worker and less efficient worker.
3. It is not easy to understand and operate.
4. High productive workers get less incentive.

6. Gantt - Plan

- This plan is based on the careful time and motion study. A standard time is fixed for doing a particular task, workers actual performance is compared with standard time and his efficiency is determined.
- If a worker takes more time than the standard time to complete the task (i.e. his efficiency is below 100%), he is given wages for the time taken by him or guaranteed minimum wage, and if a worker taken standard time to complete the job (i.e.100% efficiency) he is given a wages of standard time and a bonus of 20% on the wages earned (or a bonus of some percentage of the time taken).
- If a worker completes the task in less than the standard time, i.e. his efficiency is more than100%, he is given wages for the standard time and a bonus of 20 % of his normal earnings.
- Thus for every reduction in time the plan ensures progressive increasing total wages. For this reason, the plan is also known as “Progressive Rate System”.

Advantages:

1. The plan is not harsh as Taylor’s differential piece rate system, so it is more acceptable to the workers.
2. It is simple to understand and operate.
3. It ensures guaranteed time wages to the workers who are below average.
4. It makes distinction between efficient and inefficient workers.
5. Fixed cost per unit decreases with increase in production due to incentive for efficient workers.

Disadvantages:

1. Like Taylor’s Differential piece rate system, it divides workers in to two competing categories one who earns the bonus and the other who doesn’t earn the bonus. This brings disunity among workers and become unacceptable to the labor union.
2. The guaranteed time wage may not encourage efficiency if workers feel satisfied with the time wages